Onkar Jadhav

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Summary

Final-year CS student (9.08 CGPA, PCCOE) with hands-on experience building production AI systems: a hierarchical deep learning equity platform covering all 100 Nifty stocks, an open-source LLM governance benchmark evaluating bias across US, India, and EU legal frameworks, and an agentic wellness platform with multi-modal input and deterministic health analytics. Comfortable owning the full stack from model architecture to deployed UI. Looking for a role where I can ship systems that matter from day one.

Education

Bachelor of Technology in Computer Science *Nov 2022 – Jun 2026*
Pimpri Chinchwad College of Engineering, Savitribai Phule Pune University CGPA: 9.08 (up to 7th semester)

- Relevant Coursework: Machine Learning, Deep Learning, NLP, Data Structures & Algorithms, Database Systems, Operating Systems

Engineering Experience

StockXpert

Nov 2025 – Present

Independent Research Project | Python, TensorFlow, FastAPI, Next.js, TypeScript, Tailwind CSS, Cloudflare R2 GitHub Live

- Architected a hierarchical deep learning system combining ResNet, GRU, Bi-LSTM, and multi-head attention to model cross-horizon price dependencies across all **100 Nifty NSE equities**; ensemble design achieved a **12% improvement in short-horizon return correlation** over single-architecture baselines, with reduced directional error on high-volatility stocks.
- Constructed a leakage-free feature pipeline from raw OHLCV data incorporating RSI, MACD, ADX, Bollinger Bands, SMA/EMA distance, and volatility indicators with strict temporal train/validation/test splits — enabling clean walk-forward validation with zero lookahead bias across the full 100-stock universe.
- Designed a multi-objective Huber loss with directional regularisation and tail-event oversampling, improving signal quality on low-liquidity and high-volatility equities where standard MSE loss degrades.
- Built a snapshot-first serving layer on Cloudflare R2 with market-aware TTL caching, delivering **sub-second API latency across all 100 Nifty stocks** during market hours while eliminating redundant inference cost.
- Delivered a full-stack trading dashboard (Next.js + FastAPI) with live stock screener, real-time alerts, OHLCV candlestick charts, technical indicator overlays, and 1D–10D forecast tables with calibrated confidence scores — serving as the primary interface for daily forecast consumption across the full Nifty universe.

GovBench

Mar 2026 – Present

AI Safety Research | Python, LLM Evaluation Pipelines, Sentiment & Hallucination Detection, Adversarial Testing GitHub Live

- Designed and open-sourced an LLM evaluation framework assessing model readiness for judicial and governmental deployment across **6 pillars**: demographic bias, procedural integrity, corruption resistance, jurisdictional awareness, transparency, and minority protection — pillars chosen because MMLU-style benchmarks expose none of these failure modes.
- Implemented controlled demographic isolation testing across **12 identity variants** per scenario with naturalistic prompting, measuring inherent bias in bail, sentencing, welfare, and immigration decisions across **US, India, and EU** legal systems.
- Evaluated 4 production LLMs (Claude Sonnet, Gemini Flash, DeepSeek, GPT-4o) across baseline, pressure, and adversarial modes; top performers scored **90%**+ while exposing concrete failures — minority protection (GPT-4o: **63%**), jurisdictional awareness (DeepSeek: **56.7%**).
- Built an automated scoring pipeline combining sentiment variance analysis, position drift tracking, and hallucination detection to generate composite deployment-readiness grades, enabling unattended evaluation on new model releases.

Stride

2026 – Present

Product Engineering | Next.js, Convex, OpenRouter, Agentic AI, Multi-modal LLM, Docker GitHub Live

- Built natural-language workout parsing that converts free-text input (e.g. "30 min run, 3x10 squats at 60kg") into structured sets/reps/duration/MET values — eliminating rigid input templates and enabling accurate calorie computation without LLM arithmetic.
- Designed a four-component deterministic TDEE engine (BMR via Mifflin-St Jeor, MET-based NEAT from occupation and lifestyle, EAT from workouts, TEF) that handles all health computation while LLMs handle only intent extraction — preventing opaque model arithmetic errors in production calorie and macro calculations.
- Architected a multi-modal, multi-agent wellness platform with five specialised domain agents (diet, workout, sleep, hydration, habits) sharing persistent behavioral memory, adaptive analytics, and multi-modal input (text, voice, image, barcode).
- Designed adaptive UX driven by behavioral signals: casual users see streamlined logging, advanced users see detailed analytics, overwhelmed users see simplified interfaces — all from the same codebase without manual user-managed preferences.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Java
Frontend: React.js, Next.js, Tailwind CSS, HTML/CSS, Electron
Backend: FastAPI, Node.js, Express.js, BullMQ, Celery, Redis, REST APIs
AI / ML: TensorFlow, PyTorch, RAG Pipelines, pgvector, BM25/RRF, LLM Evaluation, Agentic AI, Multi-modal AI, XGBoost, HuggingFace, OpenRouter
LLM Providers: Anthropic Claude, OpenAI GPT, Gemini, Groq, Deepgram, whisper.cpp
Databases: PostgreSQL (+ pgvector), MongoDB, Convex
Cloud & DevOps: AWS, Cloudflare R2, Docker, Git, GitHub, GitHub Actions (CI/CD)